

French Cheese Shop

AI in Requirements Engineering

A French cheese shop exposes how AI systems must interpret vague requests, depend on domain knowledge, and be evaluated under uncertainty.

Juho Vepsäläinen

6.5.2026

**Ambiguity
Data
Evaluation**



Photo: jackmac34 / Pixabay

Learning Outcomes

1

Interpret vague requests

See why a short sentence can hide extra requirements the system still needs.

Learning Outcomes

1

Interpret vague requests

See why a short sentence can hide extra requirements the system still needs.

2

Treat data as artifact

See why domain data and product knowledge become part of the requirements.

Learning Outcomes

1

Interpret vague requests

See why a short sentence can hide extra requirements the system still needs.

2

Treat data as artifact

See why domain data and product knowledge become part of the requirements.

3

Evaluate uncertainty

Judge AI by fit, trust, and usefulness, not only by right or wrong answers.

Customer Query

A Vague Customer Request

“

I want something
like Brie,
but stronger.

Easy for a shopkeeper,
hard for a machine.

Photo: lee_2 / Pixabay

Customer Query

A Vague Customer Request

“

I want something
like Brie,
but stronger.

Easy for a shopkeeper,
hard for a machine.

Photo: lee_2 / Pixabay

Similarity

Texture, milk type,
region, serving context

Customer Query

A Vague Customer Request

“

I want something
like Brie,
but stronger.

Easy for a shopkeeper,
hard for a machine.

Photo: lee_2 / Pixabay

Similarity

Texture, milk type,
region, serving context

Strength

Age, aroma,
pungency, salt

Customer Query

A Vague Customer Request

“

I want something
like Brie,
but stronger.

Easy for a shopkeeper,
hard for a machine.

Photo: lee_2 / Pixabay

Similarity

Texture, milk type,
region, serving context

Strength

Age, aroma,
pungency, salt

Constraints

Budget, stock,
allergies, pairings

A Vague Customer Request

“

I want something
like Brie,
but stronger.

Easy for a shopkeeper,
hard for a machine.

Photo: lee_2 / Pixabay

Similarity

Texture, milk type,
region, serving context

Strength

Age, aroma,
pungency, salt

Constraints

Budget, stock,
allergies, pairings

Good answer

Best match, shortlist,
or clear explanation

Conceptual Shift

Traditional RE vs AI-Supported RE

The key shift is from building fixed features to guiding behavior when the system must interpret uncertain input.

Traditional

- Search cheeses by type
- Validate fields and stock
- Same input, same output
- Success = feature works

Conceptual Shift

Traditional RE vs AI-Supported RE

The key shift is from building fixed features to guiding behavior when the system must interpret uncertain input.

Traditional

- Search cheeses by type
- Validate fields and stock
- Same input, same output
- Success = feature works

AI-augmented

- Recommend from vague language
- Use data and domain knowledge
- Outputs depend on interpretation
- Success = useful, trusted fit

Conceptual Shift

Traditional RE vs AI-Supported RE

The key shift is from building fixed features to guiding behavior when the system must interpret uncertain input.

Traditional

- Search cheeses by type
- Validate fields and stock
- Same input, same output
- Success = feature works

AI-augmented

- Recommend from vague language
- Use data and domain knowledge
- Outputs depend on interpretation
- Success = useful, trusted fit

Requirements are no longer just functions. They also define how the system interprets, learns, and is evaluated.

Challenge 1

Challenge 1: Hidden Requirements

One vague request hides several requirements that the system must make explicit.

Observed request

“Something like
Brie,
but stronger.”

People infer context.
Software needs it stated.

What does the system still need to know?

Challenge 1

Challenge 1: Hidden Requirements

One vague request hides several requirements that the system must make explicit.

Observed request

“Something like
Brie,
but stronger.”

People infer context.
Software needs it stated.

What does the system still need to know?

Like

texture, milk type,
serving context

Challenge 1

Challenge 1: Hidden Requirements

One vague request hides several requirements that the system must make explicit.

Observed request

“Something like
Brie,
but stronger.”

People infer context.
Software needs it stated.

What does the system still need to know?

Like

texture, milk type,
serving context

Stronger

age, aroma,
pungency, salt

Challenge 1

Challenge 1: Hidden Requirements

One vague request hides several requirements that the system must make explicit.

Observed request

“Something like
Brie,
but stronger.”

People infer context.
Software needs it stated.

What does the system still need to know?

Like

texture, milk type,
serving context

Stronger

age, aroma,
pungency, salt

Good fit

budget, stock,
profile, explanation

Source: [2] Kiyavitskaya et al. (2008)

Challenge 1

Challenge 1: Hidden Requirements

One vague request hides several requirements that the system must make explicit.

Observed request

“Something like
Brie,
but stronger.”

People infer context.
Software needs it stated.

What does the system still need to know?

Like

texture, milk type,
serving context

Stronger

age, aroma,
pungency, salt

Good fit

budget, stock,
profile, explanation

AI systems need these meanings made explicit and testable.

Source: [2] Kiyavitskaya et al. (2008)

Challenge 2

Challenge 2: Data Requirements

The requirements also include the product, interaction, and domain data behind the prompt.

What data does the system need?

Challenge 2

Challenge 2: Data Requirements

The requirements also include the product, interaction, and domain data behind the prompt.

What data does the system need?

Product catalog

Milk, region, age,
texture, rind, stock

Challenge 2

Challenge 2: Data Requirements

The requirements also include the product, interaction, and domain data behind the prompt.

What data does the system need?

Product catalog

Milk, region, age,
texture, rind, stock

Interaction data

Accepted choices,
rejections, profiles

Challenge 2

Challenge 2: Data Requirements

The requirements also include the product, interaction, and domain data behind the prompt.

What data does the system need?

Product catalog

Milk, region, age,
texture, rind, stock

Interaction data

Accepted choices,
rejections, profiles

Shop knowledge

Common advice
and pairings

Challenge 2

Challenge 2: Data Requirements

The requirements also include the product, interaction, and domain data behind the prompt.

What data does the system need?

Product catalog

Milk, region, age,
texture, rind, stock

Interaction data

Accepted choices,
rejections, profiles

Shop knowledge

Common advice
and pairings

Data is not just input. It shapes what the system can know and recommend.

Source: [3] Ahmad et al. (2023)

Challenge 3

Evaluation Under Uncertainty

Evaluation is not only about whether the feature runs, but whether the result is useful, trusted, and good enough.

Where could the recommendation fail?

Challenge 3

Evaluation Under Uncertainty

Evaluation is not only about whether the feature runs, but whether the result is useful, trusted, and good enough.

Where could the recommendation fail?

User input

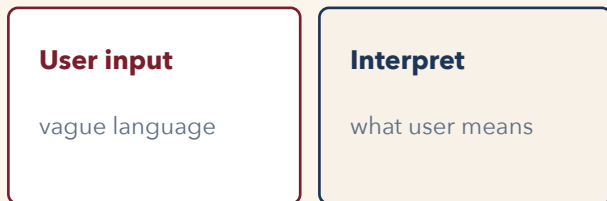
vague language

Challenge 3

Evaluation Under Uncertainty

Evaluation is not only about whether the feature runs, but whether the result is useful, trusted, and good enough.

Where could the recommendation fail?

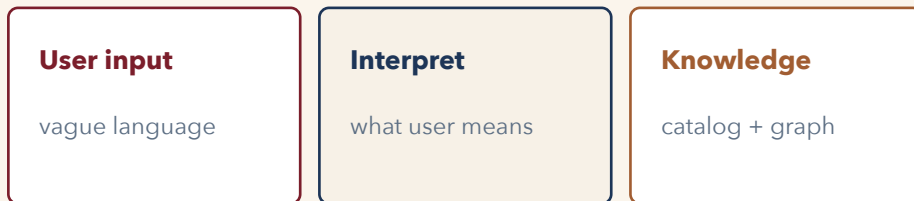


Challenge 3

Evaluation Under Uncertainty

Evaluation is not only about whether the feature runs, but whether the result is useful, trusted, and good enough.

Where could the recommendation fail?



Challenge 3

Evaluation Under Uncertainty

Evaluation is not only about whether the feature runs, but whether the result is useful, trusted, and good enough.

Where could the recommendation fail?

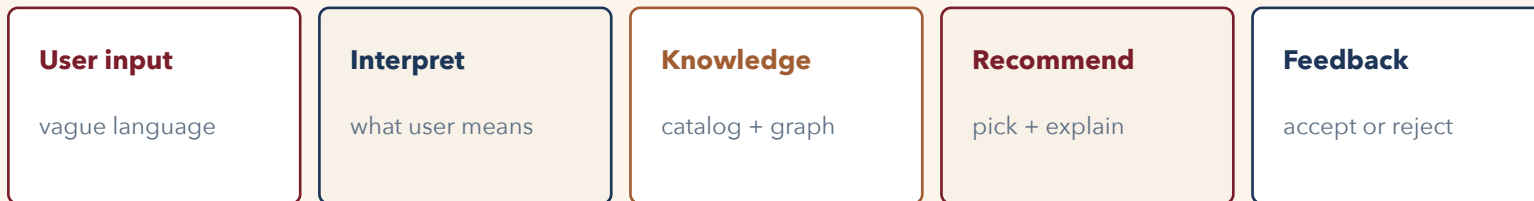


Challenge 3

Evaluation Under Uncertainty

Evaluation is not only about whether the feature runs, but whether the result is useful, trusted, and good enough.

Where could the recommendation fail?



Main Claim

AI Changes RE

AI changes requirements engineering
from specifying system functions
to shaping behavior, learning, and evaluation.

Questions and Comments

References

References

Cited sources

[1]

B. Nuseibeh and S. Easterbrook. "Requirements Engineering: A Roadmap." In The Future of Software Engineering, ICSE 2000.

[2]

N. Kiyavitskaya, N. Zeni, L. Mich, and D. M. Berry. "Requirements for tools for ambiguity identification and measurement in natural language requirements specifications." Requirements Engineering, 13(3), 2008.

[3]

K. Ahmad, M. Abdelrazek, C. Arora, M. Bano, and J. Grundy. "Requirements engineering for artificial intelligence systems: A systematic mapping study." Information and Software Technology, 158, 2023.

Architecture Decision Records

Here they work as living design notes that capture key choices for an uncertain system.

Record key assumptions

Write down the system goals, checks, and limits, not only the chosen tool.

Connect goals to code

Link quality goals to prompts, ranking logic, and test criteria.

Keep it testable

Acceptable means it meets today's tests, not that it is always correct.

Example design record

Context: request is vague
Decision: semantic search + reranking
Evaluation: useful and explainable
Checks: stock and domain fit

Good enough for the current shop context.